# Qianli Shen

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## Education

**National University of Singapore**, Singapore — Aug. 2025
**Doctor of Philosophy(Ph.D.)** in Computer Science — Advisor: Kenji Kawaguchi

**Peking University**, China — July 2020
**Bachelor of Science(B.S.)** in Computer Science — Advisor: Zhanxing Zhu

## Experience

**Senior Research Engineer** at Alibaba Tongyi Lab, Hangzhou — Sept. 2025 - Present
Data Infra & Pipeline for Qwen, Agentic Data Engineering

**LLM Engineer Intern** at Baichuan Inc., Beijing — July. 2024 - Dec. 2024
Baichuan-M1 Pretraining/Reasoning

## Selected Publications (* indicates equal contribution)

[1] **Baichuan-m1: Pushing the medical capability of large language models**
Baichuan Team
*Tech Report*

[2] **BOTS: A Unified Framework for Bayesian Online Task Selection in LLM Reinforcement Finetuning**
**Qianli Shen**, Daoyuan Chen, Yilun Huang, Zhenqing Ling, Yaliang Li, Bolin Ding, Jingren Zhou
*International Conference on Learning Representations (ICLR), 2026*

[3] **Memory-Efficient Gradient Unrolling for Large-Scale Bi-level Optimization**
**Qianli Shen**, Yezhen Wang, Zhouhao Yang, Xiang Li, Haonan Wang, Yang Zhang, Jonathon Scarlett, Zhanxing Zhu, Kenji Kawaguchi
*In Advances in Neural Information Processing Systems (NeurIPS), 2024*

[4] **The Stronger the Diffusion Model, the Easier the Backdoor: Data Poisoning to Induce Copyright Breaches Without Adjusting Finetuning Pipeline**
Haonan Wang, **Qianli Shen**, Yao Tong, Yang Zhang, Kenji Kawaguchi
*BUGS Workshop @ In Advances in Neural Information Processing Systems (NeurIPS), 2023 (Oral)*
*In International Conference on Machine Learning (ICML), 2024 (Oral)*

[5] **VA3: Virtually Assured Amplification Attack on Probabilistic Copyright Protection for Text-to-Image Generative Models**
Xiang Li*, **Qianli Shen***, Kenji Kawaguchi
*In Computer Vision and Pattern Recognition (CVPR), 2024 (Highlight)*

[6] **PICProp: Physics-Informed Confidence Propagation for Uncertainty Quantification**
**Qianli Shen**, Wai Hoh Tang, Zhun Deng, Apostolos Psaros, Kenji Kawaguchi
*In Advances in Neural Information Processing Systems (NeurIPS), 2023*

[7] **Deep Reinforcement Learning with Robust and Smooth Policy**
**Qianli Shen***, Yan Li*, Haoming Jiang, Zhaoran Wang, Tuo Zhao
*In International Conference on Machine Learning (ICML), 2020*

## Softwares

[1] **Data-Juicer Agents: Towards Agentic Data Processing**
https://github.com/datajuicer/data-juicer-agents

[2] **Data Juicer: The Data Operating System for the Foundation Model Era**
https://github.com/datajuicer/data-juicer